DiaTool-DPO

Multi-Turn Direct Preference Optimization for Tool-Augmented LLMs

Enhancing TA-LLM Dialogue via Automatic Preference Pair Construction · ArXiv 2504.02882

Models TA-LLM dialogue as a 5-state Markov Decision Process

Automatically constructs chosen/rejected trajectory pairs — no human labeling

Achieves near-GPT-4o conversational tool use on open-source models

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

Why supervised fine-tuning is not enough for tool-augmented dialogue

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Fails to Ask Follow-up Questions

When a user query is missing required arguments, SFT models proceed to call tools without gathering missing slot values, producing incorrect or incomplete results.

Hallucinates Tool Invocations

Without negative examples, SFT models learn to call tools even when arguments are unavailable — generating plausible-looking but fabricated parameter values.

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Fails to Reject Out-of-Scope Calls

SFT models cannot reliably identify when no suitable tool exists, leading to spurious tool calls that should have been rejected with an explanatory message.

DiaTool-DPO addresses these gaps by training on explicit chosen vs. rejected dialogue trajectories via Direct Preference Optimization.

Modeling TA-LLM Dialogue as a 5-State Markov Decision Process

Each conversation turn is a state transition — enabling rigorous trajectory comparison

S1

Initial State

No history. Out-of-scope → rejection; otherwise transition to S2 or S3.

S2

Tool Selected w/o Complete Slots

Tool identified but required parameters missing. Slot-filling Q&A repeats here.

S3

Tool Selected w/ Complete Slots

Both tool selection and all argument values are determined. Ready to call.

S4

Wait for Tool Response

Tool call executed. Awaiting external tool results.

S5

Complete

Tool results received. Final completion message generated for user.

State Trajectories by Query Type

Type 1 (Complete query)

S1 → S3 → S4 → S5

Type 2 (Incomplete query)

S1 → S2×N → S3 → S4 → S5

Type 3 (Out-of-scope)

S1 → S1

Three Query Types Define Distinct State Trajectories

Query classification drives which preference pairs are constructed during training

Query Type	State Trajectory	Description	Example Scenario
Type 1 (Complete)	S1 → S3 → S4 → S5	Query contains all required arguments. No slot-filling needed. Direct tool invocation.	"Find restaurants near Times Square open after 9pm" — location and time both provided.
Type 2 (Incomplete)	S1 → S2×N → S3 → S4 → S5	Query lacks some arguments. Slot-filling Q&A repeats N times until all required fields are gathered.	"Book a flight to Paris" — destination given but date, class, and passenger count missing.
Type 3 (Out-of-scope)	S1 → S1	Requested functionality is unavailable. No matching tool exists. Model must reject with explanation.	"What is the weather on Mars?" — no suitable tool available; must return a rejection message.

The three-type taxonomy enables automatic construction of meaningful preference pairs: each type has a well-defined correct trajectory that can be contrasted with common error patterns.

Automatic Paired Trajectory Construction Without Human Labels

10 preference pair categories covering all major dialogue failure modes

Preference Pair Logic

Chosen = correct dialogue trajectory

Rejected = common error trajectory

Automatic generation — no human annotation required

3 learning lessons across 10 pair categories:

① Prevent redundant slot-filling (Type 1)

② Prevent slot hallucination (Type 2)

③ Correct tool call accept/reject (Type 3)

Query	Chosen Traj.	Rejected Traj.	Lesson	Count	Diff.
T1	1→3→4→5	1→2→3→4→5	No redundant slots	2,089	Easy
T1	1→3→4→5	1→(2×N)→3→4→5	No redundant slots	562	Hard
T1	1→3→4→5	1→(2×M)→3→4→5	No redundant slots	2,530	Hard
T1	1→3→4→5	1→1	Tool call accept	2,090/562	E/H
T2	1→2→3→4→5	1→3→4→5	No slot hallucin.	2,089	Easy
T2	1→(2×N)→..	1→3→4→5	No slot hallucin.	562	Hard
T2	1→(2×N)→..	1→(2×M)→..	No slot hallucin.	2,530	Hard
T2	—	1→1	Tool call accept	2,089/562	E/H
T3	1→1	1→3→4	Tool call reject	567	Hard
T3	1→1	1→(2×N)→3→4	Tool call reject	562	Hard

SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

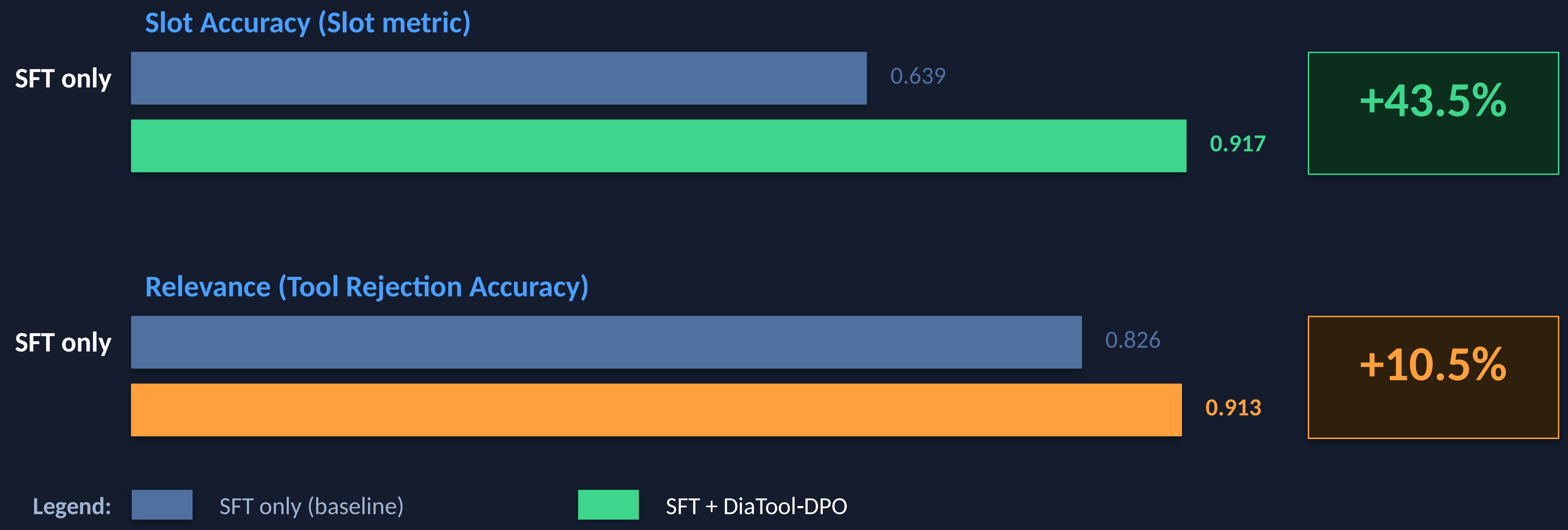
FunctionChat-Bench evaluation — Call, Completion, Slot, Relevance, Micro Avg, Macro Avg

Model	Method	Call	Completion	Slot	Relevance	Micro Avg	Macro Avg
Prop.-8B	DiaTool-DPO only	0.314	0.700	**0.833**	0.609	0.575	0.614
Prop.-8B	SFT only	**0.900**	0.916	0.694	**0.913**	0.870	0.856
Prop.-8B	SFT + DPO	0.886	**0.929**	**0.833**	0.826	**0.884**	**0.868**
Prop.-3.1B	DiaTool-DPO only	0.357	0.551	0.528	0.391	0.455	0.457
Prop.-3.1B	SFT only	**0.771**	0.817	0.750	**0.826**	**0.790**	0.791
Prop.-3.1B	SFT + DPO	0.743	**0.871**	**0.833**	**0.826**	0.765	**0.818**
LLaMA-3-8B	DiaTool-DPO only	0.029	0.449	0.056	0.261	0.205	0.199
LLaMA-3-8B	SFT only	0.843	**0.957**	0.639	0.826	0.844	0.816
LLaMA-3-8B	SFT + DPO	**0.857**	0.929	**0.917**	**0.913**	**0.905**	**0.904**

Bold gold = best value per model group | Evaluated on FunctionChat-Bench

LLaMA-3-8B: +43.5% Slot Accuracy and +10.5% Relevance with DiaTool-DPO

The largest open-source model gains — DiaTool-DPO pushes near GPT-4o performance



Full LLaMA-3-8B results: SFT + DiaTool-DPO

Call: 0.843→0.857 | Completion: 0.957→0.929 | Slot: 0.639→0.917 | Relevance: 0.826→0.913 | Micro Avg: 0.844→0.905 | Macro Avg: 0.816→0.904

Approaches near GPT-4o performance: 94.8% in information gathering, 91% in tool call rejection

Key Takeaways: RL-Based Dialogue Control for Production TA-LLMs

DiaTool-DPO — a principled, scalable path to reliable tool-augmented conversation

01 **MDP Formulation Enables Systematic Pairing**

5 internal states + 3 query types fully enumerate dialogue failure modes, making preference pair construction rigorous and exhaustive rather than ad hoc.

02 **No Human Annotation Required**

Chosen/rejected trajectory pairs are generated automatically from the MDP structure, eliminating the expensive human labeling bottleneck common in RLHF pipelines.

03 **DPO Complements — Does Not Replace — SFT**

DiaTool-DPO alone underperforms significantly. The correct recipe is SFT first for task grounding, then DPO to refine slot-filling and rejection behavior.

04 **Slot-Filling and Rejection See Largest Gains**

LLaMA-3-8B: Slot +43.5% (0.639→0.917) and Relevance +10.5% (0.826→0.913). Macro Avg rises to 0.904 — near GPT-4o conversational tool-use performance.

05 **Language-Agnostic and Production-Ready**

Primary experiments in Korean; method generalizes to English. Opens a clear path to GPT-4o-level dialogue control on smaller open-source models in production.